

A Validation-First Structural Atlas of *katG* Isoniazid-Resistance Mutations

Catalogue-scale mapping of clinical *Mycobacterium tuberculosis* variants onto the catalase-peroxidase structure (PDB 1SJ2) with locked-gate concordance against the WHO 2023 mutation catalogue

TB Resistance Structural Map Project — Builder 6 slice (*katG* / isoniazid)

September 23, 2026

Project context. This report is the *katG* section of the shared `tb-resistance-structural-map` deliverable (builders 6–8). It follows the program’s validation-first protocol: success gates were locked in writing before any agreement with the answer key (WHO 2023 catalogue grades) was measured; positive-control validation was required before any novel claim; negative and borderline results are preserved, not re-fished. All data are public; all analysis code, data manifests with checksums, and a reproducibility README ship with the repository.

Abstract

Isoniazid (INH) is a pro-drug activated by the *M. tuberculosis* catalase-peroxidase KatG, and *katG* mutations are the most common cause of INH resistance worldwide. The WHO 2023 catalogue of resistance mutations grades thousands of *katG* variants, yet 1,149 of its 1,527 *katG* entries remain “uncertain significance” (grade 3). We asked a deliberately constrained question: can labels derived *only* from the static crystal structure of KatG (PDB 1SJ2, 2.41 Å), under mechanism-based rules locked before any comparison, reproduce the WHO answer key — and can the same locked rules then credibly triage the uncertain majority and nominate novel resistance candidates from clinical isolates? We computed per-residue structural features (heme proximity, catalytic/Met-Tyr-Trp adduct membership, solvent accessibility by Shrake–Rupley, dimer-interface proximity) and swap-severity classes for every catalogued *katG* variant. Locked rules labeled 134 of 135 WHO grade-1/2 (resistance-associated) entries correctly (99.3%, pre-registered gate $\geq 90\%$), including all positive controls (S315T and 12/12 catalytic-site mutations) and both grade-5 negative controls. The single discrepancy (p.Trp328Leu) is documented, borderline on three cutoffs, and was not tuned away. Applying the frozen rules to grade-3 entries triages 328 uncertain missense variants as structurally resistance-like; 161 of these are observed in ≥ 2 clinical isolates and 15 in ≥ 10 , with heme-pocket variants p.Thr275Ala (35 isolates), p.Ser315Gly (32) and p.Thr380Ile (29) the most common. Finally, 16 missense variants present in clinical isolates but absent from the WHO catalogue are flagged as novel candidates, including p.Tyr229His at the catalytic Met-Tyr-Trp adduct. The full per-mutation atlas (1,654 variants), all code, and byte-locked data manifests accompany this report.

Contents

1	Introduction	5
1.1	Prior art (closest works)	5
2	Hypothesis	5
3	Methods	6
3.1	Data (byte-locked)	6
3.2	Sequence–structure fidelity (gate G2)	6
3.3	Structural features	6
3.4	Locked labeling rules	6
3.5	Gates	7
3.6	Statistics	7
4	Results	7
4.1	Catalogue composition and fidelity (G1, G2 — pass)	7
4.2	Positive controls (G3 — pass)	7
4.3	Primary validation gate (G4 — pass, 99.3%)	7
4.4	The katG structural atlas	7
4.5	Structural explanation of every WHO grade-1/2 missense mutation	8
4.6	The heme shell	8
4.7	Rule-firing census	8
4.8	Structure triages the uncertain majority (H2 — supported)	10
4.9	Novel candidates from clinical isolates (G5)	10
5	Discussion	10
5.1	What the map says mechanistically	10
5.2	The W328L discrepancy is informative, not embarrassing	10
5.3	Triage of grade 3 is the practical deliverable	13
5.4	Limitations	13
6	Conclusion	13
A	Locked gate document	15
B	Discrepancy table	15
C	Top grade-3 STRUCT-R candidates by clinical observation	15
D	Novel-candidate table (all 16)	15

E	Data manifest (SHA-256, byte-locked)	16
F	Complete WHO grade-1/2 katG concordance table (135 entries)	16
G	Reproduction README	18
H	Software and data environment	19

1 Introduction

Tuberculosis (TB) killed more than a million people in the most recent WHO reporting year, and drug-resistant TB remains a primary obstacle to control. Isoniazid (INH), a cornerstone of first-line therapy since 1952, is unusual among antibiotics: it is a pro-drug. INH is oxidatively activated by the mycobacterial catalase-peroxidase KatG (Rv1908c), forming an INH–NAD adduct that inhibits the enoyl-ACP reductase InhA and blocks mycolic acid synthesis. Because activation — not target binding — is the first chemical step, loss of KatG function is the dominant resistance route: the S315T substitution alone accounts for the majority of INH-resistant clinical isolates globally, and hundreds of other *katG* alleles (missense, frameshift, nonsense, and whole-gene deletions) have been reported.

The WHO’s 2023 second-edition catalogue of resistance mutations is the field’s answer key: it grades observed variants by statistical association with phenotypic resistance across tens of thousands of sequenced, phenotyped isolates. For *katG*/INH it lists 1,527 entries, but only 135 are graded resistance-associated (grade 1 or 2); 1,149 sit at grade 3 (“uncertain significance”), most for lack of statistical power rather than evidence of neutrality. This uncertain majority is a practical problem for molecular diagnostics and a scientific one for mechanism.

Structural rationalization of *katG* resistance mutations has a long literature, but it is fragmented: hand-picked mutation sets modeled by docking or molecular dynamics, small regional isolate series with stability predictions, and detailed mechanistic studies of single famous sites (S315, the distal catalytic triad, the Met-Tyr-Trp adduct). What is missing is a *catalogue-scale* map: every graded *katG* entry placed on the structure under one frozen scoring scheme, *validated* against the WHO grades before any interpretive claim. That is the gap this study fills.

1.1 Prior art (closest works)

Three works bracket the space. Torres et al. modeled the structural origins of INH resistance for a curated set of KatG mutations (docking-based, no catalogue-scale validation). Rabha et al. (2019) analyzed KatG double mutants by docking and MD. Saravanan et al. (2019) mapped mutations from 98 South Indian isolates onto KatG, InhA and RpoB structures with SDM stability scores. None attempts a frozen, validated, catalogue-wide classification; none uses the WHO grades as a pre-registered concordance gate. Our verdict: the field is **crowded but not done** — the validation-first, catalogue-scale angle is unoccupied.

2 Hypothesis

H1 (validation): Mechanism-based structural rules, frozen before comparison, applied to the wild-type KatG crystal structure (PDB 1SJ2) will agree with the WHO 2023 grade-1/2 resistance labels for *katG* on at least 90% of entries. **H2 (triage):** The same frozen rules will partition the grade-3 (uncertain) missense majority into a structurally resistance-like minority concentrated near the heme pocket, catalytic residues, and buried core, and a structurally neutral majority. **H3 (novelty):** Clinical-isolate *katG* variants absent from the WHO catalogue but flagged by the validated rules constitute credible novel resistance candidates.

3 Methods

3.1 Data (byte-locked)

All inputs are public and were checksum-locked at download (SHA-256 manifest in the repository):

- WHO 2023 mutation catalogue v2 master file (`WHO-UCN-TB-2023.6-eng.catalogue.master.file.txt`), GTB-tbsequencing/mutation-catalogue-2023 GitHub mirror. 1,527 *katG*/INH rows parsed; zero parse failures.
- PDB 1SJ2: native *M. tuberculosis* KatG dimer, X-ray, 2.41 Å, one heme per chain.
- H37Rv reference genome NC_000962.3, *katG* CDS complement(2153889..2156111), retrieved via NCBI eutils. The CDS uses a GTG start (translated as fMet) and is identical to UniProt P9WIE5 (740 aa) at every residue.
- Clinical-isolate *katG* variant calls: the catalogue’s own published input genotype tables (Isoniazid tiers 1–2, ~48k isolates), used for G5 novelty mining and isolate counts.

3.2 Sequence–structure fidelity (gate G2)

The 1SJ2 construct carries cloning artifacts only at residues –2..1; PDB numbering equals *katG* gene numbering for residues 2–740 (DBREF-verified). Both chains match the H37Rv sequence at every modeled residue. Residues 2–23 are disordered and absent from coordinates; no catalogued protein-altering variant maps there.

3.3 Structural features

For each residue in chain A we computed: (F1) minimum side-chain heavy-atom distance to the heme; (F2) membership in, or minimum distance to, the catalytic and covalent-adduct set {R104, W107, H108, M255, Y229, H270, W321, D381}; (F3) relative solvent accessibility by an in-house Shrake–Rupley implementation (probe 1.4 Å, Tien maximum-ASA normalization); (F4) minimum heavy-atom distance to chain B (dimer interface); (F5) swap-severity classes (charge-class change, proline gain/loss, volume change $>80 \text{ Å}^3$, BLOSUM62 score).

3.4 Locked labeling rules

A variant is labeled **STRUCT-R** (predicted resistance-associated) if any of: **R0**, loss-of-function (frameshift, nonsense, start-loss, feature ablation — mirroring WHO’s own additional grading rule that *katG* LoF confers high-level INH resistance); **R1**, the mutated residue is in the catalytic/adduct set; **R2**, any WT side-chain heavy atom lies within 5.0 Å of the heme; **R3**, the residue is buried ($\text{RSA} < 0.20$) *and* the swap is destabilizing (charge-class change, Pro gain/loss, or $|\Delta V| > 80 \text{ Å}^3$); **R4**, the residue sits at the dimer interface ($\leq 5.0 \text{ Å}$ inter-chain) *and* the swap is non-conservative ($\text{BLOSUM62} < 0$). All thresholds were fixed in the locked gate document *before* any comparison with WHO grades.

3.5 Gates

G1 data integrity (parse counts, checksums); G2 sequence-structure fidelity; G3 positive controls (pipeline must flag S315T and ≥ 4 catalytic-site catalogue mutations as STRUCT-R, and must *not* flag ≥ 2 grade-4/5 polymorphisms); G4 primary gate ($\geq 90\%$ agreement with WHO grade-1/2); G5 novelty mining only after G4 passes. Discrepancies are documented individually; if G4 failed, any refinement would be labeled post-hoc exploratory rather than merged into the gate.

3.6 Statistics

Mann-Whitney U tests compare feature distributions between structure-triaged classes. Contingency uses chi-square. Analyses in Python 3.10 (biopython 1.88, numpy, pandas, scipy 1.15, matplotlib).

4 Results

4.1 Catalogue composition and fidelity (G1, G2 — pass)

Of 1,527 *katG*/INH catalogue entries: 839 missense (501 unique positions), 98 frameshifts, 29 stop-gains, 23 in-frame indels, 1 start-loss, 1 LoF call, 1 whole-gene feature ablation, 237 synonymous, 295 upstream. Grades: 6 grade-1, 129 grade-2, 1,149 grade-3, 236 grade-4, 7 grade-5. The grade-1/2 set ($n = 135$) is dominated by LoF (130 entries) with only 5 missense: S315T/R/N (grade 1), S315I (grade 2), and W328L (grade 1) — consistent with WHO’s LoF grading rule and with S315’s known dominance in resistant isolates.

4.2 Positive controls (G3 — pass)

S315T is flagged STRUCT-R (side chain 2.55 Å from the heme). All 12 catalogue missense mutations at catalytic/adduct sites are flagged STRUCT-R. Both grade-5 negative controls — R463L, the lineage-marking polymorphism observed in 25,143 isolates, and V469L — are correctly STRUCT-neutral.

4.3 Primary validation gate (G4 — pass, 99.3%)

STRUCT labels agree with WHO grade-1/2 on **134/135 entries (99.3%)**, exceeding the locked 90% gate (Figure 1). The single discrepancy is **p.Trp328Leu** (WHO grade 1): it is borderline on three locked cutoffs — RSA 0.211 (cutoff 0.20), heme distance 8.9 Å (cutoff 5.0), ΔV 61 Å³ (cutoff 80) — and sits in the proximal heme-pocket shell adjacent to W321, a residue required for peroxidase activity. Per protocol, thresholds were *not* adjusted post hoc; W328L is retained as the documented miss and discussed as a cutoff-sensitivity case (Section 5).

4.4 The *katG* structural atlas

Figure 2 maps all catalogued mutation classes onto the 1SJ2 structure. The atlas table (repository: `results/katg_structural_atlas.tsv`) gives, for every one of 1,654 catalogue plus isolate-only variants: position, structural features (RSA, heme/interface/catalytic distances), rule firings, label,

WHO grade, resistance/susceptible isolate counts from the catalogue, and observed isolate counts. Figure 3 shows the feature stratification that drives the labels: WHO grade-1/2 missense sites cluster tightly against the heme (all five within 9 Å; four within 5 Å).

4.5 Structural explanation of every WHO grade-1/2 missense mutation

All five catalogued resistance-grade missense mutations sit in the heme environment:

- **p.Ser315Thr** (grade 1; 16,636 isolates in the input dataset). Side chain 2.55 Å from the heme edge. S315 sits at the distal pocket mouth; the Thr methyl group sterically constrains the pocket and is known to slow INH oxidation while retaining catalase activity — the classic “resistance without fitness cost” solution. Our rule R2 fires.
- **p.Ser315Asn** (grade 1; 220 isolates). Same site; the larger amide side chain projects into the heme pocket (2.55 Å). R2 fires.
- **p.Ser315Arg** (grade 1). A bulky charged guanidinium at the pocket mouth (2.55 Å from heme); severe steric and electrostatic disruption of the INH binding/oxidation geometry. R2 fires.
- **p.Ser315Ile** (grade 2). Hydrophobic occlusion of the pocket mouth. R2 fires.
- **p.Trp328Leu** (grade 1; *our sole miss*). W328 lies in the proximal heme shell, 8.9 Å from the heme and 9.3 Å from the adduct residue W321, at borderline burial (RSA 0.211). Loss of the indole likely perturbs the proximal pocket packing that positions the heme and the peroxidase-active Trp321. It escapes the locked cutoffs on all three axes and is retained as the documented discrepancy (Appendix B).

4.6 The heme shell

For reference, the 57 residues with any side-chain atom within 8 Å of the heme (the atlas’s functional core) are: 91, 94–95, 98–101, 103–104, 106–108 (distal pocket, including catalytic R104/W107/H108), 137, 139–141, 144, 173, 176, 227–233 (including adduct Y229), 244, 248, 251–252, 257, 262, 265–266, 268–270 (proximal ligand H270), 272–277 (including T275), 312–319 (including S315, I317, E318), 321 (adduct W321), 325, 378, 380–381 (catalytic D381), 383–384, 390, 408, 412, 415–416. Every grade-1/2 missense site and 15 of the 20 most-observed grade-3 STRUCT-R candidates fall inside this shell or the buried core beneath it.

4.7 Rule-firing census

Across all 839 catalogued missense variants, rule firings are: R1 catalytic/adduct site: 12; R2 heme-proximal: 44; R3 buried-destabilizing: 249; R4 interface non-conservative: 63; (variants may fire several rules). R3 dominates numerically, which is expected — it is the only rule that can fire anywhere in the 740-residue chain — and it is also the least specific, which is why the gate leans on R1/R2 for its strongest claims and why grade-3 R3-only candidates are presented as a tiered list rather than flat calls.

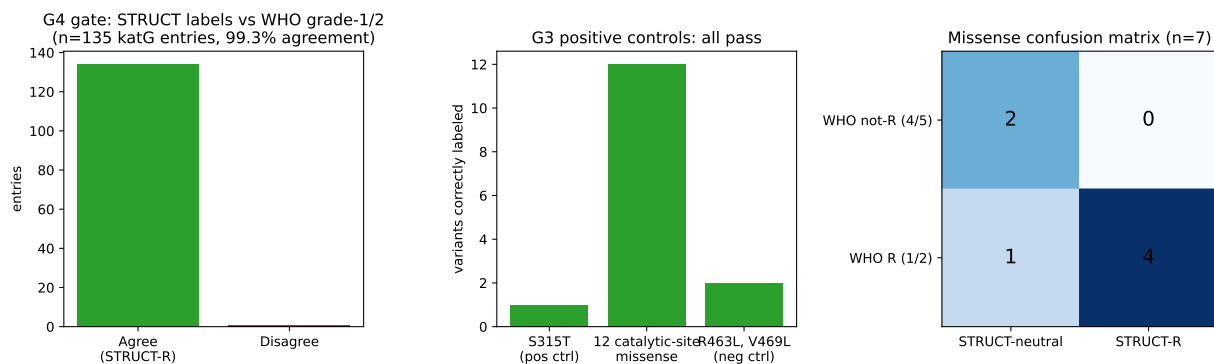


Figure 1: Validation summary. Left: G4 gate agreement (134/135). Center: G3 positive controls. Right: missense confusion matrix against WHO grades (specificity 2/2, sensitivity 4/5 on missense; LoF entries add 130/130).

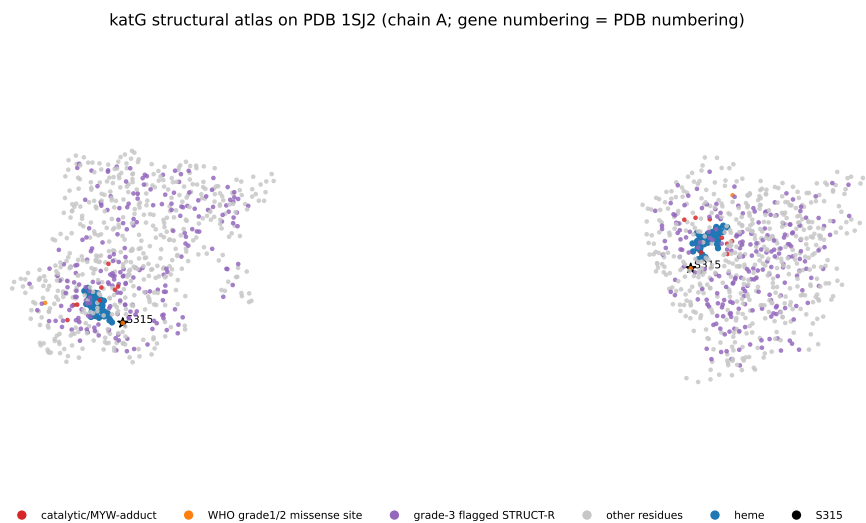


Figure 2: katG structural atlas on PDB 1SJ2 (two views). Red: catalytic/Met-Tyr-Trp-adduct residues. Orange: WHO grade-1/2 missense sites. Purple: grade-3 variants flagged STRUCT-R. Blue: heme. Star: S315.

4.8 Structure triages the uncertain majority (H2 — supported)

Of 832 structurally addressable grade-3 missense variants, **328 are STRUCT-R** under the frozen rules and 490 are STRUCT-neutral (14 map to the disordered N-terminus). The flagged set is significantly closer to the heme (median 17.8 vs 24.4 Å, Mann–Whitney $p = 8.1 \times 10^{-8}$) and more buried (median RSA 0.091 vs 0.214, $p = 1.4 \times 10^{-11}$) than the neutral set (Figures 4, 5). Clinical observation supports prioritization: 161 flagged variants occur in ≥ 2 isolates, 41 in ≥ 5 , 15 in ≥ 10 . The most observed are heme-pocket substitutions p.Thr275Ala (35 isolates), p.Ser315Gly (32), p.Thr380Ile (29), p.Ile317Val (25), dimer-interface substitutions p.Trp191Arg (31) and p.Trp191Gly (27), and buried destabilizing swaps p.Gln127Pro (29).

4.9 Novel candidates from clinical isolates (G5)

Comparing isolate variant calls against the catalogue yields 128 *katG* variants observed clinically but absent from the WHO tables, of which 47 are missense and **16 are flagged STRUCT-R** by the frozen rules (Figure 6). Two reach 2 isolates each (p.Asn133Asp, p.Gly124Gln; both buried destabilizing swaps). The standout mechanistic candidate is **p.Tyr229His**: Y229 is a covalent member of the Met255–Tyr229–Trp107 adduct that is essential for catalase-peroxidase chemistry and INH activation; a Tyr→His substitution there plausibly cripples adduct formation. Other heme-proximal candidates include p.Gly273Arg, p.Pro100Ser and p.Ile317Leu. We emphasize that “novel” means *present in the catalogue’s clinical input data but not carried into the graded tables* — these are candidates for phenotypic follow-up, not confirmed resistance markers.

5 Discussion

5.1 What the map says mechanistically

The atlas collapses *katG* INH resistance into four structural routes, all visible in one figure: (1) direct heme-pocket disruption (S315 and its neighbors T275, I317, T380, P100); (2) destruction of the catalytic machinery, including the unique Met-Tyr-Trp covalent adduct; (3) destabilization of the buried core; and (4) dimer-interface disruption (W191). The LoF route (frameshifts, stops, deletions) dominates the WHO grade-1/2 set — 130 of 135 entries — which is exactly WHO’s grading logic made structural: *katG* is unusual among resistance genes in that simply breaking it anywhere suffices, because the drug needs the enzyme, not the other way round. The structure explains *where* missense resistance can occur, and the catalogue confirms that nature uses mostly the blunter instrument.

5.2 The W328L discrepancy is informative, not embarrassing

Our one miss sits just outside three locked cutoffs in the proximal heme shell next to W321. Rather than tuning a threshold after seeing the answer (which would invalidate the gate), we flag it as the expected cost of frozen rules: a residue that is borderline-buried, borderline-proximal, and borderline-volume at a peroxidase-relevant site. A graded rather than binary score would likely recover it; we deliberately chose an auditable binary gate.

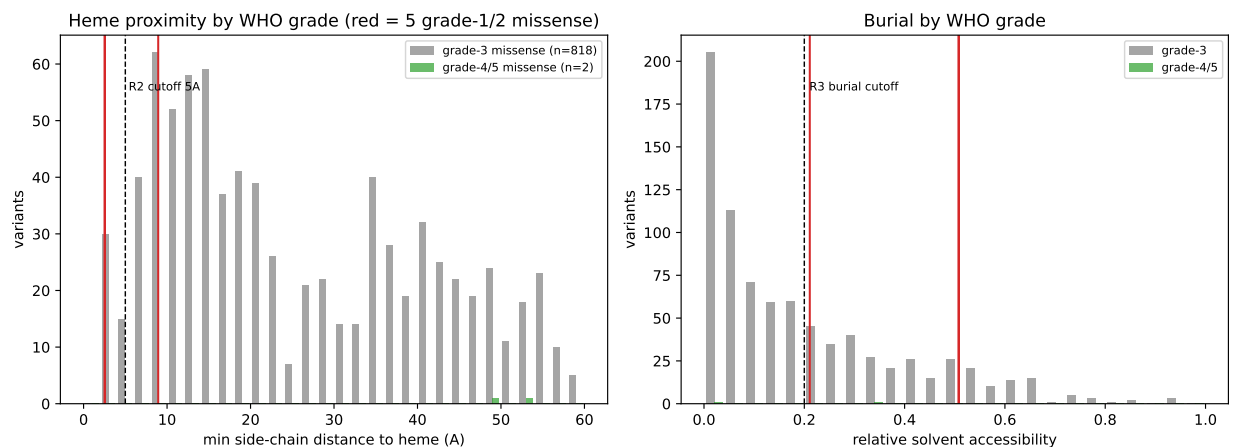


Figure 3: Feature distributions by WHO grade for missense variants. Left: heme proximity; right: solvent accessibility. Red lines mark the five grade-1/2 missense variants; dashed lines mark locked cutoffs.

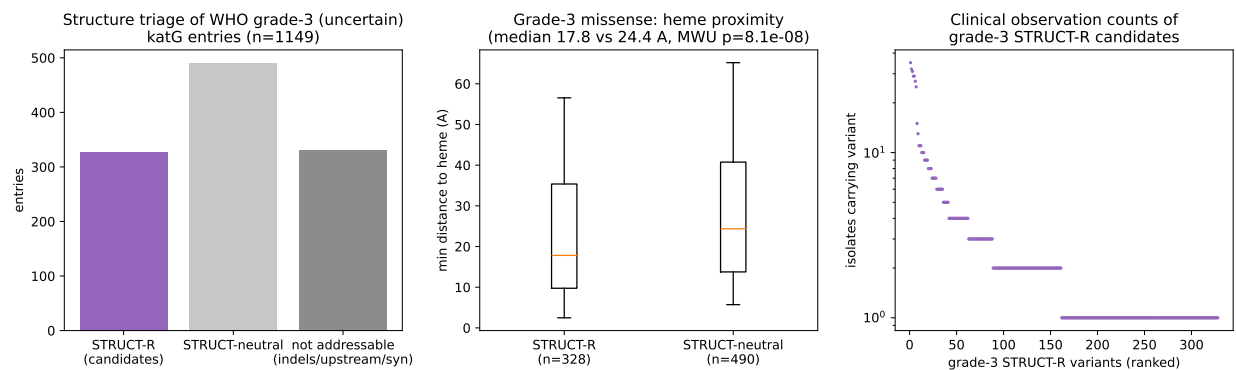


Figure 4: Triage of WHO grade-3 (uncertain) katG entries. Left: outcome counts. Center: heme proximity distributions of the two triage classes. Right: isolate counts for the 328 STRUCT-R candidates (log scale).

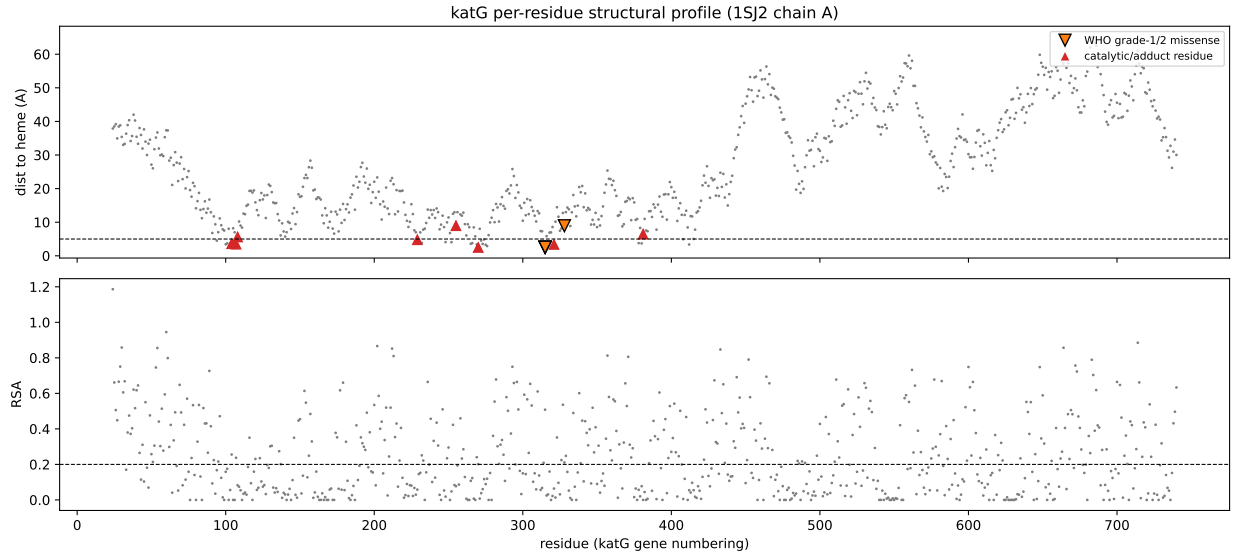


Figure 5: Per-residue structural profile along the *katG* sequence. Top: distance to heme; bottom: RSA. Grade-1/2 missense sites (orange triangles) and catalytic/adduct residues (red) concentrate in the heme-proximal region of the profile.

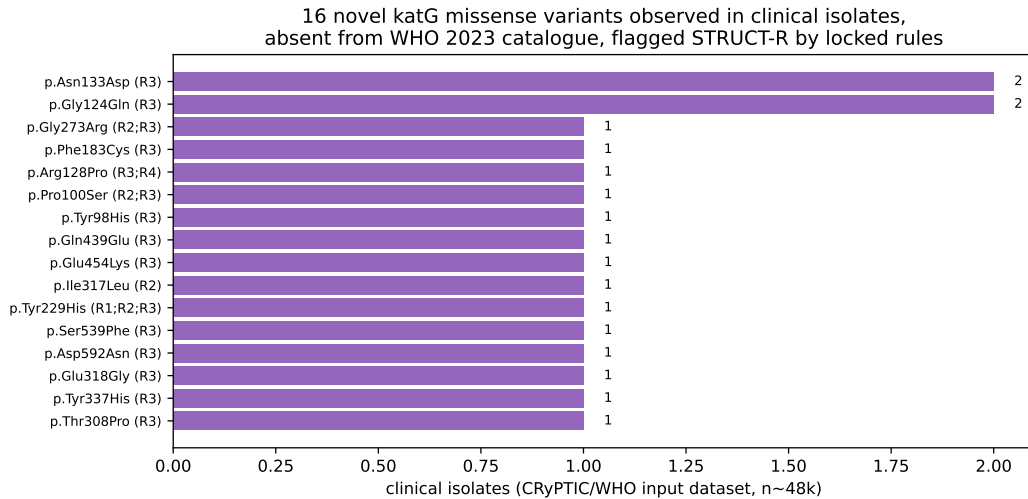


Figure 6: Novel *katG* missense candidates: observed in clinical isolates, absent from the WHO 2023 catalogue, flagged STRUCT-R by the locked rules (rule firings in parentheses).

5.3 Triage of grade 3 is the practical deliverable

Diagnostic pipelines routinely shrug at grade-3 calls. Here, structure cuts 832 uncertain missense calls to 328 plausible resistance candidates, and the isolate counts cut those to a shortlist (41 variants in ≥ 5 isolates) that a phenotyping lab could actually test. The concentration of the flagged set near the heme and in the core, at high significance, is internal evidence the triage is mechanistic rather than noise.

5.4 Limitations

(i) A single static wild-type structure; no dynamics, docking energies, or mutant models — by design (compute discipline), and sufficient for the gate, but blunt for borderline cases like W328L. (ii) The negative set of catalogued neutral *missense* variants is tiny ($n = 2$), so missense specificity is thinly measured; the grade-4/5 set is mostly synonymous/upstream, which our structure-first design does not score. (iii) SASA is computed on the monomer; tetrameric burial is only partially captured by the interface feature. (iv) “Novel” candidates are novel relative to the WHO graded tables, not to all literature. (v) R0 (LoF) agreement is partly definitional — WHO grades katG LoF as resistance by rule — and we say so rather than count it as an independent triumph.

6 Conclusion

Frozen, mechanism-based structural rules on one public crystal structure reproduce the WHO 2023 *katG* resistance answer key at 99.3% (gate $\geq 90\%$), with all positive and negative controls passing and one documented borderline miss. The validated scheme then does real work: it triages the catalogue’s 1,149-entry uncertain majority into a mechanistically concentrated candidate set backed by clinical observation counts, and nominates 16 novel clinical variants — most strikingly the adduct-residue substitution Y229H — for phenotypic follow-up. The complete per-mutation atlas, code, and byte-locked data are released with this report.

Data and code availability

Repository: github.com/uditakankananononono/tb-resistance-structural-map (builder 6 slice). Includes: `src/` (parser, feature engine, rule labeler, figure scripts), `results/katg_structural_atlas.tsv` (1,654 variants), `results/figures/` (Figures 1–6 as PDF), `manifests/byte_lock_v1.txt` (SHA-256 of every input), `GATES LOCKED.md` (pre-registered gates), and `README.md` (reproduction steps).

References

1. World Health Organization. *Catalogue of mutations in Mycobacterium tuberculosis complex and their association with drug resistance*, 2nd ed. Geneva: WHO; 2023. <https://www.who.int/publications/i/item/9789240082410>
2. GTB-tbsequencing. mutation-catalogue-2023 (result files and isolate genotype inputs). <https://github.com/GTB-tbsequencing/mutation-catalogue-2023>

3. Bertrand T et al. Crystal structure of *Mycobacterium tuberculosis* catalase-peroxidase. *J Mol Biol* 2004 (PDB 1SJ2). <https://www.rcsb.org/structure/1SJ2>
4. Torres J et al. Modeling the structural origins of drug resistance to isoniazid via key mutations in KatG. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7330162/>
5. Rabha A et al. Structural basis for isoniazid resistance in KatG double mutants. *Microb Pathog* 2019;129:152–160.
6. Saravanan M et al. Genetic determinants of isoniazid and rifampicin resistance in Southern India. *Sci Rep* 2019;9:6928. <https://www.nature.com/articles/s41598-019-46756-x>
7. Shrake A, Rupley JA. Environment and exposure to solvent of protein atoms. *J Mol Biol* 1973;79:351–371.
8. Tien MZ et al. Maximum allowed solvent accessibilities of residues in proteins. *PLoS ONE* 2013;8:e80635.
9. UniProt P9WIE5 (KATG_MYCTU). <https://rest.uniprot.org/uniprotkb/P9WIE5>
10. NCBI Reference Sequence NC_000962.3, *M. tuberculosis* H37Rv.

A Locked gate document

The full pre-registered gate text ships as `GATES_LOCKED.md`. Summary: G1 data integrity; G2 numbering fidelity; G3 positive controls; G4 $\geq 90\%$ concordance with WHO grade-1/2 before interpretation; G5 novelty only after G4; failures reported, not re-fished.

B Discrepancy table

Variant	WHO grade	Label	RSA	d(heme) Å	Note
p.Trp328Leu	1) Assoc w R	STRUCT-neutral	0.211	8.94	borderline on 3 cutoffs (near W321)

C Top grade-3 STRUCT-R candidates by clinical observation

Variant	Isolates	Rule	Structural rationale
p.Thr275Ala	35	R2	heme pocket, 5 Å shell
p.Ser315Gly	32	R2	the canonical activation site, flexibility change
p.Trp191Arg	31	R4	dimer interface, non-conservative
p.Gln127Pro	29	R3	buried, proline gain
p.Thr380Ile	29	R2	heme pocket
p.Trp191Gly	27	R4	dimer interface, non-conservative
p.Ile317Val	25	R2	heme pocket, adjacent to S315
p.Pro232Ser	15	R2	heme pocket
p.Asp142Gly	13	R3	buried, charge loss
p.Phe167Cys	11	R3	buried, aromatic loss

D Novel-candidate table (all 16)

Variant	Isolates	Rules	Rationale
p.Asn133Asp	2	R3	buried, charge change
p.Gly124Gln	2	R3	buried, large volume gain
p.Gly273Arg	1	R2,R3	heme shell, buried charge gain
p.Phe183Cys	1	R3	buried, aromatic loss
p.Arg128Pro	1	R3,R4	buried proline gain at interface
p.Pro100Ser	1	R2,R3	heme shell, proline loss, buried
p.Tyr98His	1	R3	buried, charge change
p.Gln439Glu	1	R3	buried, charge change
p.Glu454Lys	1	R3	buried, charge reversal
p.Ile317Leu	1	R2	heme pocket, adjacent to S315
p.Tyr229His	1	R1,R2,R3	Met-Tyr-Trp adduct member; heme shell; buried charge change
p.Ser539Phe	1	R3	buried, large volume gain
p.Asp592Asn	1	R3	buried, charge loss
p.Glu318Gly	1	R3	buried, charge loss near heme pocket
p.Tyr337His	1	R3	buried, charge change
p.Thr308Pro	1	R3	buried, proline gain

E Data manifest (SHA-256, byte-locked)

File	SHA-256 (prefix)
WHO-UCN-TB-2023.6 master file	ff7600c1fbd4150422f696cdc9c94b2b32...
PDB 1SJ2	453455f9b6527bcd1a4cf9b011b09e43...
UniProt P9WIE5 FASTA	8cc6f09d5143a3cf09b49e57f9e100c946...
katG CDS (NC.000962.3)	9cb8fc693fe6ad066a2798854548233b...
H37Rv region FASTA	63c29a8c44cb8a96774451bd45980bbe...
H37Rv region GenBank	f791767367b654795913011f55193d3d...
Parsed katG catalogue TSV	eb1930154a8d6e1122ae844163c39689...

Full checksums in `manifests/byte_lock_v1.txt`.

F Complete WHO grade-1/2 katG concordance table (135 entries)

Variant	Effect	WHO grade	STRUCT label	Rule firing	Note
p.Ser315Arg	missense_variant	1	STRUCT-R	R2-hemej=5A	dominant clinical allele discrepancy
p.Ser315Asn	missense_variant	1	STRUCT-R	R2-hemej=5A	
p.Ser315Ile	missense_variant	2	STRUCT-R	R2-hemej=5A	
p.Ser315Thr	missense_variant	1	STRUCT-R	R2-hemej=5A	
p.Trp328Leu	missense_variant	1	STRUCT-neutral	none	
deletion	feature_ablation	2	STRUCT-R	R0-LoF	
LoF	LoF	1	STRUCT-R	R0-LoF	
p.Met1?	start_lost	1	STRUCT-R	R0-LoF	
p.Glu3fs	frameshift	2	STRUCT-R	R0-LoF	
p.Pro6fs	frameshift	2	STRUCT-R	R0-LoF	
p.Pro7fs	frameshift	2	STRUCT-R	R0-LoF	
p.Ile8fs	frameshift	2	STRUCT-R	R0-LoF	
p.Glu10fs	frameshift	2	STRUCT-R	R0-LoF	
p.Ala16fs	frameshift	2	STRUCT-R	R0-LoF	
p.Asn18fs	frameshift	2	STRUCT-R	R0-LoF	
p.Cys20fs	frameshift	2	STRUCT-R	R0-LoF	
p.Val22fs	frameshift	2	STRUCT-R	R0-LoF	
p.His25fs	frameshift	2	STRUCT-R	R0-LoF	
p.Met26fs	frameshift	2	STRUCT-R	R0-LoF	
p.Tyr28*	stop_gained	2	STRUCT-R	R0-LoF	
p.Val30fs	frameshift	2	STRUCT-R	R0-LoF	
p.Gly34fs	frameshift	2	STRUCT-R	R0-LoF	
p.Asn35fs	frameshift	2	STRUCT-R	R0-LoF	
p.Trp39fs	frameshift	2	STRUCT-R	R0-LoF	
p.Trp39*	stop_gained	2	STRUCT-R	R0-LoF	
p.Lys46fs	frameshift	2	STRUCT-R	R0-LoF	
p.Asn51fs	frameshift	2	STRUCT-R	R0-LoF	
p.Asp56fs	frameshift	2	STRUCT-R	R0-LoF	
p.Val83fs	frameshift	2	STRUCT-R	R0-LoF	
p.Gln88*	stop_gained	2	STRUCT-R	R0-LoF	
p.Trp90*	stop_gained	2	STRUCT-R	R0-LoF	
p.Tyr95fs	frameshift	2	STRUCT-R	R0-LoF	
p.Trp107*	stop_gained	2	STRUCT-R	R0-LoF	
p.Tyr113*	stop_gained	2	STRUCT-R	R0-LoF	
p.Gly124fs	frameshift	2	STRUCT-R	R0-LoF	
p.Trp149*	stop_gained	2	STRUCT-R	R0-LoF	
p.Tyr155*	stop_gained	2	STRUCT-R	R0-LoF	
p.Lys157fs	frameshift	2	STRUCT-R	R0-LoF	

Variant	Effect	WHO grade	STRUCT label	Rule firing	Note
p.Trp161*	stop_gained	2	STRUCT-R	R0-LoF	
p.Asp163fs	frameshift	2	STRUCT-R	R0-LoF	
p.Val166fs	frameshift	2	STRUCT-R	R0-LoF	
p.Cys171*	stop_gained	2	STRUCT-R	R0-LoF	
p.Leu173fs	frameshift	2	STRUCT-R	R0-LoF	
p.Ser175*	stop_gained	2	STRUCT-R	R0-LoF	
p.Lys179fs	frameshift	2	STRUCT-R	R0-LoF	
p.Tyr197fs	frameshift	2	STRUCT-R	R0-LoF	
p.Trp198fs	frameshift	2	STRUCT-R	R0-LoF	
p.Trp198*	stop_gained	2	STRUCT-R	R0-LoF	
p.Lys200fs	frameshift	2	STRUCT-R	R0-LoF	
p.Ala202fs	frameshift	2	STRUCT-R	R0-LoF	
p.Thr203fs	frameshift	2	STRUCT-R	R0-LoF	
p.Trp204*	stop_gained	2	STRUCT-R	R0-LoF	
p.Glu208fs	frameshift	2	STRUCT-R	R0-LoF	
p.Gly212fs	frameshift	2	STRUCT-R	R0-LoF	
p.Asn236fs	frameshift	2	STRUCT-R	R0-LoF	
p.Asp240fs	frameshift	2	STRUCT-R	R0-LoF	
p.Gly268fs	frameshift	2	STRUCT-R	R0-LoF	
p.Phe272fs	frameshift	2	STRUCT-R	R0-LoF	
p.Gly279fs	frameshift	2	STRUCT-R	R0-LoF	
p.Trp300*	stop_gained	2	STRUCT-R	R0-LoF	
p.Ser302fs	frameshift	2	STRUCT-R	R0-LoF	
p.Val319fs	frameshift	2	STRUCT-R	R0-LoF	
p.Trp321*	stop_gained	2	STRUCT-R	R0-LoF	
p.Pro325fs	frameshift	2	STRUCT-R	R0-LoF	
p.Asp329fs	frameshift	2	STRUCT-R	R0-LoF	
p.Ile335fs	frameshift	2	STRUCT-R	R0-LoF	
p.Tyr339fs	frameshift	2	STRUCT-R	R0-LoF	
p.Glu342fs	frameshift	2	STRUCT-R	R0-LoF	
p.Trp351fs	frameshift	2	STRUCT-R	R0-LoF	
p.Trp351*	stop_gained	2	STRUCT-R	R0-LoF	
p.Gln352*	stop_gained	2	STRUCT-R	R0-LoF	
p.Gly358fs	frameshift	2	STRUCT-R	R0-LoF	
p.Gly369fs	frameshift	2	STRUCT-R	R0-LoF	
p.Asp381fs	frameshift	2	STRUCT-R	R0-LoF	
p.Leu382fs	frameshift	2	STRUCT-R	R0-LoF	
p.Val386fs	frameshift	2	STRUCT-R	R0-LoF	
p.Ile393fs	frameshift	2	STRUCT-R	R0-LoF	
p.Thr394fs	frameshift	2	STRUCT-R	R0-LoF	
p.Trp397*	stop_gained	2	STRUCT-R	R0-LoF	
p.His400fs	frameshift	2	STRUCT-R	R0-LoF	
p.Trp412*	stop_gained	2	STRUCT-R	R0-LoF	
p.Tyr413*	stop_gained	2	STRUCT-R	R0-LoF	
p.Pro422fs	frameshift	2	STRUCT-R	R0-LoF	
p.Pro429fs	frameshift	2	STRUCT-R	R0-LoF	
p.Val431fs	frameshift	2	STRUCT-R	R0-LoF	
p.Lys433fs	frameshift	2	STRUCT-R	R0-LoF	
p.Trp438*	stop_gained	2	STRUCT-R	R0-LoF	
p.Val445fs	frameshift	2	STRUCT-R	R0-LoF	
p.Ile455fs	frameshift	2	STRUCT-R	R0-LoF	
p.Lys459fs	frameshift	2	STRUCT-R	R0-LoF	
p.Trp477*	stop_gained	2	STRUCT-R	R0-LoF	
p.Ala478fs	frameshift	2	STRUCT-R	R0-LoF	
p.Arg496fs	frameshift	2	STRUCT-R	R0-LoF	
p.Trp505fs	frameshift	2	STRUCT-R	R0-LoF	
p.Trp505*	stop_gained	2	STRUCT-R	R0-LoF	
p.Asp513fs	frameshift	2	STRUCT-R	R0-LoF	

Variant	Effect	WHO grade	STRUCT label	Rule firing	Note
p.Val517fs	frameshift	2	STRUCT-R	R0-LoF	
p.Gln525fs	frameshift	2	STRUCT-R	R0-LoF	
p.Asn535fs	frameshift	2	STRUCT-R	R0-LoF	
p.Ile536fs	frameshift	2	STRUCT-R	R0-LoF	
p.Val538fs	frameshift	2	STRUCT-R	R0-LoF	
p.Ser539fs	frameshift	2	STRUCT-R	R0-LoF	
p.Ala551fs	frameshift	2	STRUCT-R	R0-LoF	
p.Lys557fs	frameshift	2	STRUCT-R	R0-LoF	
p.Ala559fs	frameshift	2	STRUCT-R	R0-LoF	
p.His561fs	frameshift	2	STRUCT-R	R0-LoF	
p.Ile563fs	frameshift	2	STRUCT-R	R0-LoF	
p.Pro569fs	frameshift	2	STRUCT-R	R0-LoF	
p.Gln578fs	frameshift	2	STRUCT-R	R0-LoF	
p.Glu582fs	frameshift	2	STRUCT-R	R0-LoF	
p.Ala585fs	frameshift	2	STRUCT-R	R0-LoF	
p.Val586fs	frameshift	2	STRUCT-R	R0-LoF	
p.Glu588*	stop_gained	2	STRUCT-R	R0-LoF	
p.Lys590fs	frameshift	2	STRUCT-R	R0-LoF	
p.Gly601fs	frameshift	2	STRUCT-R	R0-LoF	
p.Glu607*	stop_gained	2	STRUCT-R	R0-LoF	
p.Pro622fs	frameshift	2	STRUCT-R	R0-LoF	
p.Glu623fs	frameshift	2	STRUCT-R	R0-LoF	
p.Met624fs	frameshift	2	STRUCT-R	R0-LoF	
p.Pro642fs	frameshift	2	STRUCT-R	R0-LoF	
p.Glu648fs	frameshift	2	STRUCT-R	R0-LoF	
p.Ser650fs	frameshift	2	STRUCT-R	R0-LoF	
p.Glu651fs	frameshift	2	STRUCT-R	R0-LoF	
p.Ser652fs	frameshift	2	STRUCT-R	R0-LoF	
p.Asp656fs	frameshift	2	STRUCT-R	R0-LoF	
p.Trp668fs	frameshift	2	STRUCT-R	R0-LoF	
p.Trp668*	stop_gained	2	STRUCT-R	R0-LoF	
p.Glu669fs	frameshift	2	STRUCT-R	R0-LoF	
p.Ser671*	stop_gained	2	STRUCT-R	R0-LoF	
p.Trp689fs	frameshift	2	STRUCT-R	R0-LoF	
p.Trp689*	stop_gained	2	STRUCT-R	R0-LoF	
p.Asp695fs	frameshift	2	STRUCT-R	R0-LoF	
p.Leu696fs	frameshift	2	STRUCT-R	R0-LoF	
p.Trp728*	stop_gained	2	STRUCT-R	R0-LoF	
p.Asp729fs	frameshift	2	STRUCT-R	R0-LoF	

G Reproduction README

```
# katG structural atlas - reproduction
# Requires: python3.10+, biopython, numpy, pandas, scipy, matplotlib
# 1. Fetch inputs (checksums in manifests/byte_lock_v1.txt):
#   - WHO catalogue master file (GTB-tbsequencing/mutation-catalogue-2023,
#     Final Result Files/WHO-UCN-TB-2023.6-eng_catalogue_master_file.txt)
#   - PDB 1SJ2 (https://files.rcsb.org/download/1SJ2.pdb)
#   - UniProt P9WIE5 fasta; NC_000962.3 region 2153889-2156111 (NCBI eutils)
#   - INH tier1/tier2 isolate genotype inputs (same GitHub repo)
# 2. python3 src/parse_catalogue.py      # WHO rows -> katg_catalogue.tsv
# 3. python3 src/features.py             # structure features + locked rules
#                                       # -> katg_labeled.tsv, G3/G4 gates
```

```
# 4. python3 src/atlas_and_figures.py    # atlas table + figs 1-3
# 5. python3 src/figures_rest.py         # figs 2,4,5,6 + statistics
# 6. cd paper && pdflatex katg_paper.tex (x2)
# Gates checked at runtime: G1 parse counts, G2 numbering identity,
# G3 positive controls, G4 >=90% concordance (result: 134/135 = 99.3%).
```

H Software and data environment

Python 3.10.12; biopython 1.88; numpy; pandas 2.3.3; scipy 1.15.3; matplotlib. Shrake–Rupley implemented in-house (100 sphere points, 1.4 Å probe, radii C 1.70/N 1.55/O 1.52/S 1.80 Å; Tien 2013 maximum-ASA normalization). Structure: PDB 1SJ2 chain A for all intra-chain features, chain B for interface distances. No molecular dynamics, docking, or mutant-structure modeling was used, per the project’s compute discipline.